

Graduation Project Documentation

CAI4_DAT2_S3: Team 2

Overview:

- This project presents an interactive fraud analysis dashboard developed to monitor, analyze, and interpret fraudulent transaction behavior. The solution focuses on two key perspectives: transaction volume and financial impact, allowing users to explore fraud patterns through multiple dimensions such as time, customer demographics, geographic location, and merchant categories.

Dataset

- Fraudulent transactions cases dataset from Kaggle

Data Modeling

- Fact Table: Transaction(amnt, isfraud)
- Dimensional Tables:
 - Merchant(Merchant_id, Category)
 - Customer(cc_num, gender, job, city, state)
 - Calendar(Date) "user made table"
- Relationships:
 - Transaction -> Merchant (Merchant_id)
 - Transaction -> Customer (cc_num)
 - Transaction -> Calendar (Date)

Data Preparation

Main Data cleaning Steps(all tables):

- Replace errors with null
- Remove duplicate/null rows
- Remove other unwanted columns

Transactions Table:

- Extracted/Splitted column for Date and column for Time from Date_Time column

Dax Formulas

Measure	Importance
KPI 1: Fraud Transactions	
Fraud Transactions	Number of Transactions where fraud cases occurred
%Fraud Rate	Percentage of Fraud Transactions from Total Transactions
KPI 2: Fraud Amount	
Fraud Amount	Financial loss Amount due to fraud cases
%Fraud Amount Rate	Percentage of Fraud Amount from Total Amount
%MOM	Shows difference in performance with last month

Enhancements:

- [Calculated Column in Transaction Table:](#)

```

1 Time of Day =
2 VAR Period = FORMAT('Transaction'[trans_time], "AM/PM")
3 VAR TransHour = HOUR('Transaction'[trans_time])
4 RETURN
5     SWITCH(
6         TRUE(),
7         Period = "AM" && TransHour >= 5,           "Morning",
8         Period = "AM" && TransHour < 5,           "Night",
9         Period = "PM" && TransHour >= 12 && TransHour < 17, "Afternoon",
10        Period = "PM" && TransHour >= 17,         "Evening",
11        "Night"
12    )

```

- [Customer Age in Customer Table:](#)

```

1 Customer Age =
2
3 DATEDIFF ( Customer[date_of_birth], TODAY(), YEAR )

```

- [%MOM Icon Formula:](#)

```
1 MOM Trans Icon =  
2 SWITCH(  
3   TRUE(),  
4   [MOM Fraud Transactions] < 0, "●",  
5   [MOM Fraud Transactions] <= 0.3, "●",  
6   "●"  
7 )
```

Visualization

Pages Structure:

1. KPIs -> what happened?
2. Trend-> change over time
3. Drivers-> how it happened

Charts:

Page 1: Transactions – Analyzing Fraud Transactions and Fraud Rate

KPI Cards:

Total Transactions, Fraud Transactions (+ MoM indicator), % Fraud Rate

Time Analysis:

Fraud Transactions by Month (trend over time)

Fraud Transactions by Time of Day (morning, afternoon, evening, night)

Customer Analysis:

Fraud Transactions by Customer Age (bins)

Fraud Transactions by Customer Gender

Fraud Transactions by Customer Job

Location Analysis:

Fraud Transactions by State (map view)

Merchant Analysis:

Fraud Transactions by Merchant Category

Page 2: Amounts – Analyzing Fraud Amounts

KPI Cards:

Total Amount, Fraud Amount (+ MoM indicator), % Fraud Amount Rate

Time Analysis:

Fraud Amount by Month

Fraud Amount by Time of Day

Customer Analysis:

Fraud Amount by Customer Age (bins)

Fraud Amount by Customer Gender

Location Analysis:

Fraud Amount by State

Merchant Analysis:

Fraud Amount by Merchant Category

Page 3: Details Table – Detailed Fraud BreakdownAnalysis Groups:

Category-Level Analysis: Identify high-risk categories

Merchant-Level Analysis: Compare merchants by fraud activity

Performance Indicators: % Fraud Rate and % Fraud Amount Rate with visual cues

Conditional Formatting Columns:

- Fraud Transactions by Background Color (Light Red to Hot Red)
 - %Fraud Rate by Icon (Red, Yellow, Green)
 - Fraud Amount by Background Color (Light Blue to Dark Blue)
 - %Fraud Amount Rate by Icon (Red, Yellow, Green)
-

Interaction:

- Slicers by: Quarter, Month / Category / State
 - Drill Through: From Details Table to Transactions & Amounts pages
 - Tooltips: by %Fraud Rate
 - Conditional Formatting
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Theme of UI/UX Design

- Dark theme (Navy, Red) with navigation for easy interpretation, using Claude AI

Conclusion

The developed dashboard successfully delivers a comprehensive view of fraudulent activities by combining transaction-based and amount-based analysis. Through its interactive design and well-structured visualizations, users can quickly identify trends, detect anomalies, and focus on high-risk areas.

The inclusion of a detailed table further strengthens the solution by enabling in-depth analysis at a granular level. Overall, the project demonstrates how data visualization tools can transform raw transactional data into meaningful insights, supporting more effective fraud monitoring and strategic decision-making.